

静态数码管显示实验

1、实验任务

设计 8 位数码管静态显示：实现 6 位数码管从 00000000——FFFFFFFF 循环显示，每个字符显示 0.5s 后变化。

2、实验原理：

HX4S20 板采用 8 位共阳数码管，原理图查找“使用手册”图 1.1 所示的是 8 位数码扫描显示电路，其中每个数码管的 8 个段 h、g、f、e、d、c、b、a（h 是小数点）都分别连在一起接图 1.1 DIG0-DIG7，8 个数码管分别由 8 个选通信号 k1~k8 来选择接图 1.1 SEL0-SEL7。被选通的数码管显示数据，其余关闭。如在某一时刻，k3 为低电平，其余选通信号为高电平，这时仅 k3 对应的数码管显示来自段信号端的数据，而其他 7 个数码管呈现关闭状态。根据这种电路状况，如果希望在 8 个数码管显示希望的数据，就必须使得 8 个选通信号 k1~k8 分别被单独选通，同时在段信号输入口加上希望该对应数码管上显示的数据，于是随着选通信号的扫变，就能实现扫描显示的目的。

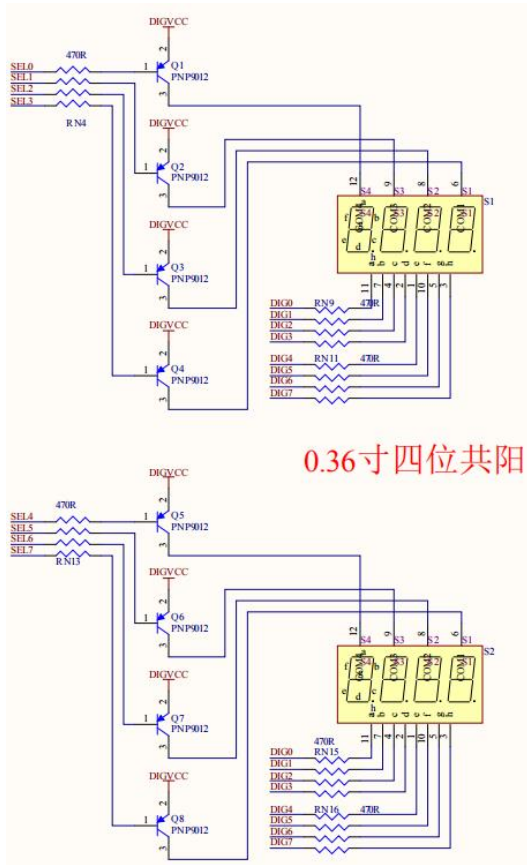


图 1.1 8 数码管连接图

3、实验程序：

数码管静态显示实验顶层文件

```

module seg_led_static_top (
    input          sys_clk ,           // 系统时钟
    input          sys_rst_n,         // 系统复位信号（低有效）
    output [5:0]    seg_sel ,          // 数码管位选
    output [7:0]    seg_led           // 数码管段选
);
//parameter define
parameter TIME_SHOW = 25'd25000_000; // 数码管变化的时间间隔 0.5s
//wire define
wire      add_flag;                  // 数码管变化的通知信号
//每隔 0.5s 产生一个时钟周期的脉冲信号
time_count #(
    .MAX_NUM    (TIME_SHOW)
) u_time_count(
    .clk         (sys_clk ),
    .rst_n       (sys_rst_n),

    .flag        (add_flag )
);
//每当脉冲信号到达时，使数码管显示的数值加 1
seg_led_static u_seg_led_static (
    .clk         (sys_clk ),
    .rst_n       (sys_rst_n),
    .add_flag    (add_flag ),
    .seg_sel     (seg_sel ),
    .seg_led     (seg_led )
);
endmodule

```

数码管静态显示模块

```

module seg_led_static (
    input          clk ,           // 时钟信号
    input          rst_n ,         // 复位信号（低有效）
    input          add_flag,       // 数码管变化的通知信号
    output reg [5:0] seg_sel ,     // 数码管位选
    output reg [7:0] seg_led      // 数码管段选
);

```

```
//reg define
reg [3:0] num;                // 数码管显示的十六进制数
//控制数码管位选信号（低电平有效），选中所有的数码管
always @ (posedge clk or negedge rst_n) begin
    if (!rst_n)
        seg_sel <= 6'b111111;
    else
        seg_sel <= 6'b000000;
end
//每次通知信号到达时，数码管显示的十六进制数值加 1
always @ (posedge clk or negedge rst_n) begin
    if (!rst_n)
        num <= 4'h0;
    else if(add_flag) begin
        if (num < 4'hf)
            num <= num + 1'b1;
        else
            num <= 4'h0;
    end
    else
        num <= num;
end
//根据数码管显示的数值，控制段选信号
always @ (posedge clk or negedge rst_n) begin
    if (!rst_n)
        seg_led <= 8'b0;
    else begin
        case (num)
            4'h0 :    seg_led <= 8'b1100_0000;
            4'h1 :    seg_led <= 8'b1111_1001;
            4'h2 :    seg_led <= 8'b1010_0100;
            4'h3 :    seg_led <= 8'b1011_0000;
            4'h4 :    seg_led <= 8'b1001_1001;
            4'h5 :    seg_led <= 8'b1001_0010;
            4'h6 :    seg_led <= 8'b1000_0010;
            4'h7 :    seg_led <= 8'b1111_1000;
            4'h8 :    seg_led <= 8'b1000_0000;
```

```

        4'h9 :    seg_led <= 8'b1001_0000;
        4'ha :    seg_led <= 8'b1000_1000;
        4'hb :    seg_led <= 8'b1000_0011;
        4'hc :    seg_led <= 8'b1100_0110;
        4'hd :    seg_led <= 8'b1010_0001;
        4'he :    seg_led <= 8'b1000_0110;
        4'hf :    seg_led <= 8'b1000_1110;
        default : seg_led <= 8'b1100_0000;
    endcase
end
end
endmodule

```

计时模块

```

module time_count(
    input          clk      ,    // 时钟信号
    input          rst_n    ,    // 复位信号
    output reg     flag      // 一个时钟周期的脉冲信号
);
//parameter define
parameter  MAX_NUM = 25000_000; // 计数器最大计数值
//reg define
reg [24:0] cnt;                // 时钟分频计数器
//计数器对时钟计数，每计时到 0.5s,输出一个时钟周期的脉冲信号
always @ (posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        flag <= 1'b0;
        cnt  <= 24'b0;
    end
    else if(cnt < MAX_NUM - 1'b1) begin
        cnt  <= cnt +1'b1;
        flag <= 1'b0;
    end
    else begin
        cnt  <= 24'b0;
        flag <= 1'b1;
    end
end

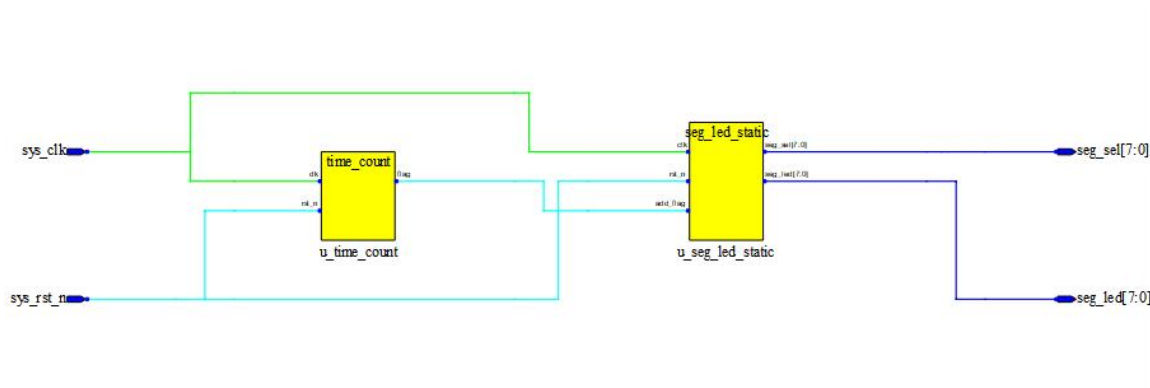
```

```

    end
end
endmodule

```

4、RTL 原理图：



5、引脚约束：

复位 rst 按键 1，输出 8 位数码管 en 接数码管使能，DATAOUT 接段选

```

set_pin_assignment { seg_led[0] }{ LOCATION = C15; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_led[1] }{ LOCATION = C16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_led[2] }{ LOCATION = B6; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_led[3] }{ LOCATION = A5; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_led[4] }{ LOCATION = B5; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_led[5] }{ LOCATION = A8; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_led[6] }{ LOCATION = A7; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_led[7] }{ LOCATION = A6; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[0] }{ LOCATION = D11; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[1] }{ LOCATION = E11; IOSTANDARD = LVCMOS33;

```

```

DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[2] } { LOCATION = F10; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[3] } { LOCATION = C13; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[4] } { LOCATION = E10; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[5] } { LOCATION = E12; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[6] } { LOCATION = E16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { seg_sel[7] } { LOCATION = F16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { sys_clk } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { sys_rst_n } { LOCATION = A2; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }

```

6、下载测试：

DIG/LED 拨上，接通数码管，拨下接通 LED，观察显示的效果

